Predictive Loss Does Not Certify Quotient Faithfulness

An Embedding-Space Account of Three Manifestations

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Abstract

Predictive criteria — bare loss, sufficient-statistic objectives, and information-bottleneck variants — measure the relationship between an encoder E and a prediction target f. Quotient faithfulness measures the relationship between E and the structure of the data-generating process $(\sim, Q, q)$. These are different relationships. The criterion is structurally silent on the second, for any encoder, regardless of whether the admissible nuisance relation $\sim$ has been declared.

This paper develops the consequences of that silence. The constructive content is the *embedding-space program*: a chain of seven objects relating the declared quotient to the latent shape the encoder constructs, with the faithfulness condition stated as a relation between them and three architectural-adequacy conditions on the latent side. The negative content is three logically independent failure modes that the criterion permits in the regime where $\sim$ has been externally declared — the easiest regime for the practitioner.

The first failure mode is geometric: the latent ambient must have at least $\mathrm{emb}_{\mathrm{top}}(Q)$ dimensions for any homeomorphic image of Q to exist inside it. Below this floor, no faithful encoding is possible. The second is graded compression: above the floor, the criterion permits encoders that approximate an embedding of a coarsening of Q rather than Q itself. The population case fixes one specific coarsening (the task map's image); the finite-sample case admits a graded simplicial family at every dimension from one up to the floor. The third is quotient-respect failure: the criterion does not enforce that equivalent inputs produce equal codes, and admits encoders faithful to quotients of $\mathcal{X}$ different from Q — including incomparable quotients that both retain nuisance distinctions and collapse lawful ones.

The unidentifiability theorem of Locatello et al. (2019) is the parallel manifestation in the regime where $\sim$ has not been declared. The three failure modes are not downstream of that result; they are facets of the same underlying problem. The paper does not claim that standard training cannot recover faithful representations. Its narrower conclusion is that loss-based evaluation, on its own, does not certify recovery. Any claim that a learned representation faithfully encodes an underlying quotient therefore requires structural information beyond the criterion — an explicit admissibility declaration, an architectural commitment that pre-fixes the embedding, or both.

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series Quotient Faithfulness in Representation Learning develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful *if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural manifestations of criterion silence prevent standard training from achieving this: the embedding floor, graded compression, and quotient-respect failure.*

Paper I *[Farkas, 2026a] (this paper): Establishes that predictive loss does not certify quotient faithfulness; develops the embedding-space program and the three structural manifestations.*

Paper II *[Farkas, 2026b]: Identifies recovery routes and admissibility sources. Establishes the graded-compression structure and its Kolmogorov complexity foundations.*

Paper III *[Farkas, 2026c]: Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding in the single-quotient setting.*

Paper IV *[Farkas, 2026d]: Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers.*

Paper V *[Farkas, 2026e]: Maps the framework against representation-learning traditions and examines fourteen bodies of experimental evidence through the embedding-space lens.*

Paper VI *[Farkas, 2026f]: Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.*

Paper VII *[Farkas, 2026g]: Empirical validation on benchmark dynamical systems (phase cylinder, Hopf fibration, double pendulum) and an industrial case study (Color Fantasy ship-hotel energy system). Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].*

Contents

Notation

Symbol	Meaning
$\mathcal{X}$	Observation space
$\sim$	Admissible nuisance relation on $\mathcal{X}$
$\mathcal{Q} = \mathcal{X}/\sim$	Lawful quotient: space of semantic states
$q : \mathcal{X} \twoheadrightarrow \mathcal{Q}$	Canonical quotient projection
Y	Target label space
$f : \mathcal{X} \to Y$	Prediction target ($f = t \circ q$)
$t : \mathcal{Q} \to Y$	Task map (factor of f through $\mathcal{Q}$)
$Z = \mathbb{R}^{D_{\text{ambient}}}$	Latent ambient
$E : \mathcal{X} \to Z$	Encoder (representation map)
$E(\mathcal{X}) \subset Z$	Latent shape
$\widetilde{E} : \mathcal{Q} \to Z$	Induced quotient representation
E_{label}	Label representation: $E_{\text{label}}(x) = f(x) = t(q(x))$
$\sim_E$	Encoder-induced relation: $x \sim_E x' \iff E(x) = E(x')$
$Q_E = \mathcal{X}/\sim_E$	Encoder-induced quotient
ϕ	Chosen homeomorphic embedding $Q \hookrightarrow \mathbb{R}^m$
$E_1 \preceq_{\text{comp}} E_2$	E_1 at least as compressed as E_2
$\dim_{\text{cov}}(\mathcal{Q})$	Covering dimension of $\mathcal{Q}$
$\text{emb}_{\text{top}}(\mathcal{Q})$	Topological embedding dimension of $\mathcal{Q}$
D_{lat}	Intrinsic dimension of the latent shape
D_{ambient}	Latent ambient dimension ($= \dim Z$, set by architecture)
S^1	Circle (canonical non-contractible quotient example)

Typographic conventions. Lowercase italic for points, labels, scalars; uppercase italic for sets, spaces, maps; calligraphic uppercase for distinguished objects ($\mathcal{X}$, $\mathcal{Q}$, $\mathcal{H}$); blackboard bold for number systems; tildes for derived versions of a base symbol. Symbols $\hookrightarrow$, $\to$, $\twoheadrightarrow$, $\cong$ denote topological embedding, continuous map, quotient (surjection), and homeomorphism respectively.

1 Introduction

1.1 The rotating object

Imagine training a neural network on a large collection of images of a single rotating object. Each image is tagged with the viewing angle, in degrees. The network sees pixels, predicts angles, and after training it achieves near-perfect test accuracy. Ask the natural question: has the network learned that these are views of the same object from different angles?

The intuitive answer is yes — if it predicts the angle correctly, surely it has learned the underlying structure. The intuitive answer is wrong, or more precisely, is not certifiably right from the training signal alone. Two representations can produce identical predictions on every training example and every held-out example, while being radically different in what they encode about the object.

One representation embeds viewing angle as a circle in the latent space: angle $0°$ sits next to $359°$, the geometry of the latent mirrors the geometry of the rotation. This is what we would want.

A second representation lays viewing angles out along a line segment, $0°$ at one end and $359°$ at the other, with no closure between them. It fits every training example perfectly while erasing the circular structure: angles at $0°$ and $359°$ are physically adjacent and geometrically distant in the latent. The criterion sees no difference between this representation and the circular one.

A third representation embeds each viewing angle along with a copy of some nuisance variable — say, lighting condition. The latent shape is a two-dimensional surface, with rotation along one axis and lighting along another. Predictions still hit the angle correctly (the readout simply ignores the lighting axis). But two images of the object at the same angle under different lighting now occupy *different* points in the latent shape, even though we declared them to be the same physical state. The criterion sees no difference between this representation and one that respects the lighting equivalence.

This paper is about why the training criterion cannot tell these three representations apart, and about the geometric floor that any of them must clear in the first place.

1.2 What counts as faithful

Call the declared equivalence structure on the data the *target quotient*. In the rotation example, the quotient is the circle S^1: distinct pixel images of the object at the same angle represent the same underlying state. In an object-recognition task, the quotient might be the set of object identities, with different lightings and viewpoints identified. In a physical system, the quotient might be an energy shell, a leaf of a foliation, or the orbit of a symmetry group acting on state space.

A learned representation is *quotient-faithful* if the induced map from the quotient into the latent space preserves enough structure for downstream use — topological structure, at minimum, and often metric structure as well. A quotient-faithful encoding of the rotation problem places the image of S^1 inside the latent space as an actual topological circle. A quotient-unfaithful encoding is the line-segment version above, the collapsed single-point version, the lighting-distinguishing surface version, or a metrically distorted version where angles that are physically close map to codes that are far apart.

Two points are worth stating up front, because they frame everything that follows.

Master claim. The predictive criterion measures the relationship between E and the prediction target f. Faithfulness is the relationship between E and the generating-process structure $(\sim, Q, q)$. These are different relationships, and the predictive criterion is structurally silent on the second relationship for any encoder. We call this property *criterion silence*, and it is the master claim of the paper.

The silence has two regimes. Without an external admissibility declaration, the criterion does not even allow faithfulness to be stated as a mathematical condition: Locatello et al. [2019] establish this for unsupervised disentanglement, and the same pattern recurs for any structural identification from loss alone. With an external admissibility declaration in hand, the criterion still admits encoders that fail every structural requirement faithfulness imposes. The three failure modes proved in this paper are the strongest form of the claim: even when the practitioner has the structural input, an unconstrained architecture, and an ideal optimiser, the criterion still does not certify faithfulness. Where the practitioner has less, the silence is broader still.

Faithfulness is geometric, not predictive. Whether a learned representation is faithful to the declared quotient is a property of the latent *geometry*, not of the predictions. Two encoders with identical predictive behaviour can have entirely different geometric relationships to the declared quotient. Faithfulness is a claim about structure; predictive accuracy is a claim about performance; the two are logically independent in ways this paper makes precise.

1.3 The chain of objects, the faithfulness condition, and the architectural conditions

To say a representation is "faithful" or "unfaithful" we need to specify *faithful to what* — and that specification involves more than one object.

A reader may ask: why introduce intermediate objects at all? The training data are discrete points; the network's parameter realisation is a finite real-valued parameter vector together with a discrete connectivity pattern. Why not connect the two directly?

The reason is that each step in the chain is forced by a different kind of constraint, and conflating them obscures where the structural questions actually live.

The first step — from the discrete training data to an underlying continuous quotient — is forced by an assumption about the world. We assume the data are samples of something simpler than themselves: a state space governed by physical law, by mathematical structure, by some generative process that does not depend on our happening to sample it at these particular points. Without this assumption, "faithfulness" has no referent, because there is nothing for the representation to be faithful to.

The second step — from the quotient to a Euclidean ambient hosting the network's representation — is forced by epistemic uncertainty and by the mechanics of computation. We rarely know the quotient's structure exactly, and even when we do, networks operate in Euclidean ambient spaces; the quotient has to be hosted somewhere the architecture can compute on it.

The third step — from the ambient to the function space the architecture realises — is forced by the structure of neural computation. The architecture's parameter count, depth, layer structure, and activation functions determine which input-to-ambient maps the network can express.

The fourth step — from the function space to the parameter-level realisation — is forced by the mechanics of how networks actually hold information. Different parameter configurations select different functions within the function space.

These distinctions justify naming a chain of objects.

The chain.

- *The quotient Q.* A compact metric space carrying the structure of interest. In the rotation example, $Q = S^1$.
- *The observation space $\mathcal{X}$.* A compact metric space on which the data live. The lawful quotient is the surjection $q : \mathcal{X} \to Q$ identifying nuisance-equivalent observations.
- *The training data $D \subset \mathcal{X}$.* A finite, discrete sample drawn from $\mathcal{X}$.
- *The latent shape $E(\mathcal{X}) \subset Z$.* The encoder's image, viewed as a topological and metric object in its own right. The encoder is defined on the full input space, not merely on the finite training data; the latent shape is a continuous geometric object, not just a finite set of points.
- *The latent ambient $Z = \mathbb{R}^{D_{\text{ambient}}}$.* The Euclidean space holding the latent shape, set by the architecture's exposed-layer width.
- *The function space $\mathcal{H}$.* The set of input-to-ambient maps the architecture can realise. Parameter count, depth, layer structure, and activation functions determine which functions are expressible.
- *The parameter realisation θ.* The finite weight configuration selecting one specific function within the function space.

The encoder relates them through

$$Q \xleftarrow{q} \mathcal{X} \xrightarrow{E} E(\mathcal{X}) \subset Z.$$

There is one substantive correspondence in this chain — the *faithfulness condition* — and three architectural conditions on whether that correspondence can be honoured.

> **The faithfulness condition.** *Equivalent observations are mapped to the same point in the latent shape, and distinct points of the declared quotient remain distinct in the shape. The topology of Q is preserved in the topology of the latent shape; in stronger settings, the metric is preserved up to bounded distortion.*

The faithfulness condition is a relationship between the world (the quotient) and the network (the latent shape). It can be satisfied only if three architectural conditions on the latent side are met:

- *Dimensional adequacy.* $D_{\text{ambient}} \geq \text{emb}_{\text{top}}(Q)$.
- *Expressive adequacy.* $\mathcal{H}$ contains a function whose induced map on Q is a topological embedding.
- *Realisation adequacy.* The parameter configuration realises such a function accessibly under the chosen training procedure.

This paper concerns dimensional adequacy. Expressive and realisation adequacy bear on which failures are realised by particular networks; the substantive treatment is forwarded to later papers in this series.

The embedding-space program. The faithfulness condition compares Q to the latent shape, but Q carries no Euclidean structure of its own. To compare them analytically we choose a homeomorphic embedding

$$\phi : Q \hookrightarrow \mathbb{R}^m, \qquad m \geq \mathrm{emb}_{\mathrm{top}}(Q),$$

called the *embedding space* of Q. Natural choices include the revolution embedding $S^n \hookrightarrow \mathbb{R}^{n+1}$ for spheres, algebraic embeddings of compact surfaces, geometric models of the eight Thurston 3-manifolds, and projective and grassmannian models for quotients with linear-algebraic structure. The embedding space is a *pure mathematical object* chosen for analytical purposes; it is not realised in the network. The network sees only the latent ambient Z.

The role of the embedding space is to act as a *bridge* between the abstract object Q and the concrete latent ambient Z. With ϕ chosen, faithfulness can be stated in Euclidean terms: the encoder is faithful when its induced map $\widetilde{E} : Q \to Z$ is, up to symmetries of Z, a C^1-close approximation of ϕ pre-composed with an isometric embedding $\mathbb{R}^m \hookrightarrow Z$. The program becomes tractable when the embedding space and the latent ambient are aligned, that is, $D_{\mathrm{ambient}} = m$. Under alignment the local well-conditioning of the approximation is supplied by the openness of embeddings in the C^1 topology: any map sufficiently C^1-close to a C^1 embedding is itself a C^1 embedding (Hirsch). This is the local well-conditioning result the program rests on; it says nothing about how to find a faithful encoder, only that — once one is found — small perturbation does not destroy faithfulness.

Criterion silence in the program. The structural problem the paper identifies is more general than the loss landscape's behaviour for any specific encoder family. The criterion supplies none of the program's structural inputs: it does not select ϕ, it does not certify alignment $D_{\mathrm{ambient}} = m$, and it does not certify that the encoder approximates ϕ rather than some other map. The three failure modes of §§3, 4, 5 are concrete manifestations of this silence in the regime where $\sim$ has been declared. They place — through the embedding-space lens above — what *kind* of encoder the criterion permits when the practitioner has done everything available short of stepping outside the criterion itself.

The silence has a further within-encoder layer worth noting. Even when all three architectural conditions are met and the criterion has selected an encoder approximating ϕ, the set of faithful encoders for fixed $(Q, \mathcal{X}, q, Z)$ is a manifold of equivalent solutions, parametrised by the symmetries of Z (translation, rotation, scaling, reflection) and by reparametrisations of the embedding within the same image. The criterion does not distinguish among members of this manifold. Two networks that have both succeeded at faithful encoding will not, in general, agree on representatives; they will agree on the homeomorphism class of the latent shape and disagree on its Euclidean placement. This is a within-task, mathematical observation; it is independent of any empirical claim about cross-modality convergence.

Architectural encoding. For specific architecture classes, the alignment $D_{\mathrm{ambient}} = m$ is not an assumption but a constructive fact. The architecture pre-commits to a specific ϕ at design time and the latent ambient is the embedding space. Group-equivariant networks [Cohen and Welling, 2016, Bronstein et al., 2021] commit to a ϕ compatible with the group action when the quotient is a G-invariant orbit space. Block-diagonal MLPs commit to a ϕ for product quotients by assigning factors to disjoint coordinate blocks. Spherical CNNs commit to a ϕ for S^2 via spherical-harmonic decompositions. Hyperbolic networks commit to a ϕ for $\mathbb{H}^n$ via the Poincaré or Lorentzian model. In each case the architecture sidesteps the search problem entirely: the embedding space is built into the layer structure, alignment is automatic, and the optimiser is left only to fit the parametrisation of ϕ, not to discover its homeomorphism class. The cost is rigidity — the architecture is correct when the declared quotient matches the design, and structurally wrong otherwise. The benefit is a problem in which Hirsch's openness applies by construction. The general-purpose case (an MLP encoder with no commitments to ϕ) is the regime in which the program's structural inputs are all unsupplied by the criterion. The three failure modes are what happens to such an encoder when the criterion is the only signal available.

1.4 Failure modes as manifestations of criterion silence

The three failure modes of §§3, 4, 5 are concrete manifestations of criterion silence in the $\sim$-declared regime. They place — through the embedding-space lens of §1.3 — what kind of encoder the criterion permits when the practitioner has declared $\sim$, allowed any architecture with adequate dimensional capacity, and allowed an ideal optimiser. The taxonomy below describes each failure mode by which embedding the encoder turns out to approximate; the structural definitions live in the sections themselves.

- *The embedding floor (§3): no embedding from Q exists.* The latent ambient is too small to host any homeomorphic image of Q; no $\phi : Q \hookrightarrow \mathbb{R}^{D_{\text{ambient}}}$ exists. The program cannot begin — there is no target embedding to approximate.
- *Graded compression (§4): the encoder approximates an embedding from a coarsening of Q.* A target embedding $\phi : Q \hookrightarrow \mathbb{R}^{D_{\text{ambient}}}$ does exist, but the criterion does not select it. At the population level, when the task map $t : Q \to Y$ is non-injective, the deterministic compression preorder selects an encoder approximating an embedding of the coarsening $t(Q)$ rather than Q itself. At the finite-sample level, the criterion admits encoders approximating embeddings of k-dimensional simplicial coarsenings of Q at every k from 1 up to the floor.
- *Quotient-respect failure (§5): the encoder approximates an embedding from a different quotient of $\mathcal{X}$.* A target embedding $\phi : Q \hookrightarrow \mathbb{R}^{D_{\text{ambient}}}$ exists, but the encoder does not factor through q. The induced map $\widetilde{E}$ is not well-defined; the latent shape is a homeomorphic image of $\mathcal{X}/{\sim_E}$, where the encoder-induced relation $\sim_E$ does not contain $\sim$. The simplest subcase is pure refinement, where $\sim_E$ is strictly finer than $\sim$ and $\mathcal{X}/{\sim_E}$ is a refinement of Q (e.g., the encoder retains nuisance variation as a latent coordinate). More generally, $\sim_E$ may be incomparable with $\sim$, and the encoder is faithful to a quotient that is structurally different from Q — both retaining nuisance distinctions and collapsing lawful ones along axes the declared quotient keeps distinct. In either subcase the encoder is faithful to a quotient of $\mathcal{X}$ on which the criterion happens to be satisfied, but which is not Q.

Three different manifestations of one structural fact: the criterion does not certify the relationship between the encoder and the generating process. In the embedding floor case, the architecture is structurally incapable of hosting any embedding of Q, and the criterion does not tell the practitioner this. In the graded compression case, the criterion actively favours an embedding from a coarsening of Q over an embedding of Q itself. In the quotient-respect case, the criterion permits embeddings from a different quotient of $\mathcal{X}$ entirely. Faithfulness to Q requires the encoder to approximate an embedding from Q itself, neither coarsened, nor refined, nor replaced by a structurally different quotient of

$\mathcal{X}$. The criterion does not enforce this. Sections 3 through 5 work each manifestation in turn; §6 collects them into the structural decomposition; §7 refines the topology-faithful condition into a metric-faithful one and shows that even when a topological embedding of Q is achieved, the criterion does not certify its geometric quality.

The unidentifiability theorem of Locatello et al. [2019] is the parallel manifestation of criterion silence in the regime where $\sim$ has not been externally declared: the criterion does not allow $\sim$ itself to be identified from data alone. The three failure modes of this paper are not downstream of the Locatello result — they are facets of the same underlying problem.

1.5 Scope and reading guide

What this paper does and does not assume. The paper does not assume that the structural identification problem is solved. The mathematical content of §§3, 4, 5 is stated in the regime where $\sim$ has been declared because that is the regime in which faithfulness can be stated as a mathematical condition; the conceptual claim is regime-independent. The framework also does not assume that Q is a smooth manifold — Q is an abstract topological space, and the results use compactness and covering dimension, which apply to general metric spaces. It does assume that the latent ambient Z is in principle capable of representing Q (this is the embedding floor of §3; when it fails, the framework is not silently presupposing it). It treats the encoder $E : \mathcal{X} \to Z$ as a black box, caring only about the map from inputs to exposed representations and not about the layers, widths, or internal bottlenecks of the network that computes it. The dimension lower bound of §3 is a constraint on $\dim Z$, not on any internal layer width.

The running example. A single running example grounds the formal results. Consider a cyclic dynamical process whose observable $x(t) \in \mathbb{R}^m$ is driven by a phase $\theta(t) \in S^1 = \mathbb{R}/2\pi\mathcal{Z}$, with an irrelevant amplitude nuisance making the observation space a cylinder $C = S^1 \times [0, 1]$. The declared quotient is $q : C \to S^1$. The prediction target is the sine projection $t(\theta) = \sin\theta$, which is non-injective: phases θ and $\pi - \theta$ produce the same output. Phase processes arise throughout science and engineering — oscillators, circadian rhythms, rotational systems, heading in navigation — and the circle $Q = S^1$ is the simplest compact topological space that is not contractible, making it the canonical test case for the three failure modes.

Forward pointer to the series. Paper I formalises the single-quotient case: one Q, one set of architectural conditions, one faithfulness condition. Readers who find this single-quotient setting too restrictive — for spaces that decompose into regions with distinct local quotient structure (atlases of manifolds, stratified spaces, mixed-regime systems) — will find the generalisation in Paper IV of this series, which formalises a quotient atlas with per-region faithfulness conditions. Paper I's Q is the single-quotient case of that framework.

Structure of the paper. §2 fixes the formal setting and the vocabulary the theorems need. §3 establishes the embedding floor. §4 establishes graded compression at the population level (Theorem 4.4) and at the finite-sample level (Theorem 4.7, Corollary 4.9). §5 establishes quotient-respect failure with both the refinement and incomparable subcases. §6 collects the three failure modes into the structural decomposition (I)–(II)–(III). §7 identifies metric distortion as a fourth, refining condition. §8 summarises and forwards open questions to the rest of the series. Appendix A collects formal definitions; Appendix B contains classical proofs; Appendix C gives theorem-by-theorem attribution; Appendix D contains the AI disclosure prompt and a worked example. The first author takes full scientific responsibility for all claims.

2 Formal Setting

This section fixes the formal vocabulary used throughout the rest of the paper. The setting is deliberately minimal: every concept used in §§3–7 is defined here. The section does not re-state the conceptual scope of the paper; that was given in §1.5.

2.1 Observation space and admissible nuisance

Let $\mathcal{X}$ be the observation space — the set of raw inputs the encoder sees. The first modelling choice is a declaration of which elements of $\mathcal{X}$ represent the same underlying state.

Definition 2.1 (Admissible nuisance relation)**.** An *admissible nuisance relation* is an equivalence relation $\sim$ on $\mathcal{X}$ that is declared or otherwise justified as identifying observations that differ only in semantically irrelevant ways. We write $x \sim x'$ when x and x' are equivalent under $\sim$, and call the equivalence classes *nuisance fibres*.

The relation $\sim$ is supplied externally, not discovered from data by any procedure described in this paper. In practice it may come from known symmetries of the domain, from physical invariances, from multiview supervision, from causal interventions, or from domain knowledge. The paper's scope is confined to consequences of having declared $\sim$; the identification problem of *discovering* $\sim$ from data is the parallel manifestation of criterion silence developed by Locatello et al. [2019], discussed in §5.7.

2.2 The lawful quotient

Definition 2.2 (Lawful quotient)**.** The pair $(\mathcal{Q}, q)$ where $\mathcal{Q} = \mathcal{X}/\sim$ and $q : \mathcal{X} \twoheadrightarrow \mathcal{Q}$ is the canonical projection sending each observation to its equivalence class is called the *lawful quotient* induced by $\sim$. The space $\mathcal{Q}$ represents the semantic structure that remains after removing nuisance variation.

For the rotation example, $\mathcal{Q} = S^1$ and q sends each image to the angle it depicts. For the cylinder running example, $\mathcal{Q} = S^1$ and q projects the cylinder $S^1 \times [0, 1]$ onto its phase coordinate.

2.3 The prediction target

Let Y be the output space and $f : \mathcal{X} \to Y$ the ground-truth labelling function.

Assumption 2.3 (Quotient-compatible target)**.** There exists a measurable map $t : \mathcal{Q} \to Y$ such that

$$f = t \circ q.$$

That is, f depends on x only through its equivalence class $q(x)$.

This is a minimal coherence condition between the nuisance declaration and the task. Note that t need not be injective. This non-injectivity of t is precisely what creates the gap between task sufficiency and quotient faithfulness that Theorem 4.4 exploits. In the cylinder running example with $t(\theta) = \sin\theta$, the points θ and $\pi - \theta$ are distinct in $\mathcal{Q} = S^1$ but map to the same label.

2.4 Representation maps and the latent ambient

Definition 2.4 (Representation map)**.** A *representation map* (or encoder) is a measurable map $E : \mathcal{X} \to Z$ where $Z = \mathbb{R}^{D_{\text{ambient}}}$ is called the *latent ambient.* The image $E(\mathcal{X}) \subset Z$ is the *latent shape.*

All structural results in this paper concern the map $E : \mathcal{X} \to Z$ as a whole and the induced map $\widetilde{E} : \mathcal{Q} \to Z$ it produces. They make no claims about the internal layer structure of any network that implements E.

2.5 Quotient-respecting representations

Definition 2.5 (Quotient-respecting representation)**.** A representation $E : \mathcal{X} \to Z$ *respects the quotient* q if there exists a measurable map $\widetilde{E} : \mathcal{Q} \to Z$ such that

$$E = \widetilde{E} \circ q.$$

The map $\widetilde{E}$ is the *induced quotient representation.*

The factorisation $E = \widetilde{E} \circ q$ means that $E(x)$ depends on x only through its equivalence class $q(x)$. Quotient respect is necessary for faithfulness but not sufficient: a constant encoder respects any quotient yet preserves no structure. §5 characterises what happens when an encoder fails to respect the quotient.

2.6 Reduction to the induced map

Observation 2.6 (Reduction)**.** Let $E : \mathcal{X} \to Z$ be a quotient-respecting representation with induced map $\widetilde{E} : \mathcal{Q} \to Z$. Then:

(a) Whether E preserves the topology of $\mathcal{Q}$ depends entirely on whether $\widetilde{E}$ is a topological embedding.

(b) Whether E preserves the metric geometry of $\mathcal{Q}$ depends entirely on whether $\widetilde{E}$ is bi-Lipschitz.

The embedding obstruction theorems are statements about $\widetilde{E} : \mathcal{Q} \to Z$, and they hold regardless of how E is computed. The expressiveness of the architecture does not help if Z cannot topologically accommodate $\mathcal{Q}$.

2.7 Codomain conventions in §4

For most of this paper the latent ambient is fixed as $Z = \mathbb{R}^{D_{\text{ambient}}}$. Throughout §4, however, representations may take values in arbitrary measurable codomains, not necessarily Euclidean. This is needed because the compression preorder of Definition 4.2 is naturally defined across codomains. The Euclidean convention resumes from §5 onward.

3 The Embedding Floor

The first manifestation of criterion silence is geometric. Once the quotient $\mathcal{Q}$ is fixed, the latent ambient must be large enough to admit a topological embedding of $\mathcal{Q}$. The mathematical content of this section is essentially definitional once $\mathrm{emb}_{\mathrm{top}}(\mathcal{Q})$ is available, supplemented by a classical covering-dimension bound. What matters is the placement: dimensional adequacy is the architectural-adequacy condition logically prior to all training-time concerns. The criterion does not certify that an architecture clears this floor.

3.1 Topology-faithful representations

Definition 3.1 (Topology-faithful representation). A continuous map $\widetilde{E} : \mathcal{Q} \to Z$ is *topology-faithful* if it is a topological embedding of $\mathcal{Q}$ onto its image: a homeomorphism from $\mathcal{Q}$ onto $\widetilde{E}(\mathcal{Q}) \subset Z$ equipped with the subspace topology.

Homeomorphism is the right criterion because it preserves all continuous topological invariants: connectedness, compactness, and local neighbourhood structure.

3.2 The embedding floor

Theorem 3.2 (Topological Embedding Floor). *Let $\widetilde{E} : \mathcal{Q} \to \mathbb{R}^{D_{\text{ambient}}}$ be topology-faithful. Then*

$$D_{\text{ambient}} \geq \text{emb}_{\text{top}}(\mathcal{Q}).$$

Proof. Definitional. See Appendix B. □

Theorem 3.3 (Covering-Dimension Floor; Brouwer 1911, Hurewicz and Wallman 1941, Engelking 1978). *Let $\mathcal{Q}$ be a compact metric space of finite covering dimension k. Then* $\text{emb}_{\text{top}}(\mathcal{Q}) \geq k$.

Proof. Direct consequence of monotonicity of covering dimension for subspaces of Euclidean space. See Appendix B. □

The covering-dimension floor is a *floor on the floor*: $\text{emb}_{\text{top}}(\mathcal{Q})$ can strictly exceed $\dim_{\text{cov}}(\mathcal{Q})$. For S^1, $\dim_{\text{cov}}(S^1) = 1$ but $\text{emb}_{\text{top}}(S^1) = 2$. The two quantities coincide for spaces that admit flat embeddings (closed intervals, Euclidean balls); they diverge for any quotient with nontrivial loop structure.

Lemma 3.4 (Quotient Embedding Requirement; Menger 1932, Kvalheim and Sontag 2024). *Let $\mathcal{Q}$ be compact Hausdorff and $\widetilde{E} : \mathcal{Q} \to \mathbb{R}^d$ continuous. If $\widetilde{E}$ is topology-faithful, then $\mathcal{Q}$ admits a topological embedding into $\mathbb{R}^d$. Equivalently: if $\mathcal{Q}$ is not embeddable in $\mathbb{R}^d$, no quotient-respecting representation with latent ambient $\mathbb{R}^d$ can be topology-faithful.*

Proof. Definitional. See Appendix B. □

Remark 3.5 (Relationship to Kvalheim and Sontag 2024). Lemma 3.4 is logically equivalent to the definition of a topological embedding and has no content beyond it. Furthermore, Kvalheim and Sontag [2024] have already proved the exact autoencoder-specific version: a perfect autoencoder with bottleneck $\mathbb{R}^k$ exists if and only if the data manifold is homeomorphic to a subset of $\mathbb{R}^k$, which directly subsumes the mathematical strength of Lemma 3.4. We restate it here not as a novel mathematical contribution, but because it forms the third and final necessary pillar of the diagnostic framework: the obstruction is on the target space, not the computation, and this needs to be stated explicitly within the three-manifestation structure. The genuinely new content in this section is the *placement* of the floor as architectural adequacy and the connection to criterion silence.

Corollary 3.6 (Below-Floor Collapse). *Let $\widetilde{E} : \mathcal{Q} \to \mathbb{R}^{D_{\text{ambient}}}$ be continuous with $D_{\text{ambient}} < \text{emb}_{\text{top}}(\mathcal{Q})$. Then $\widetilde{E}$ cannot be topology-faithful: it must either identify distinct points of $\mathcal{Q}$, fail to preserve neighbourhood topology, or both.*

Proof. Immediate contrapositive of Theorem 3.2. For compact Hausdorff $\mathcal{Q}$ and Euclidean target, a continuous injection is automatically a topological embedding, so the two failure modes collapse: non-embedding implies non-injectivity. See Appendix B. □

3.3 Entanglement: what sub-floor maps actually produce

Remark 3.7 (Entanglement, not destruction). Corollary 3.6 establishes that no topology-faithful encoding exists when $D_{\text{ambient}} < \text{emb}_{\text{top}}(\mathcal{Q})$. It does not say that the quotient structure is absent from the representation. A continuous map $\widetilde{E} : \mathcal{Q} \to Z$ with $D_{\text{ambient}} < \text{emb}_{\text{top}}(\mathcal{Q})$ still exists and will be produced by any trained encoder; what is impossible is for that map to be injective in a topologically stable way.

What actually occurs is *entanglement*: the encoder folds $\mathcal{Q}$ into the available dimensions, creating false identifications between semantically distinct states. Distinct points of $\mathcal{Q}$ must be collapsed together (the map is non-injective), and the information about which quotient class a point belongs to becomes mixed with whatever else occupies the dimensions of Z. The quotient structure is not erased — it is co-mingled with other encoded content in a way that makes it non-independently recoverable without simultaneously decoding the other entangled dimensions.

This is a different failure mode from graded compression (§4), where the criterion actively rewards collapsing $\mathcal{Q}$. The embedding floor is a geometric impossibility that operates regardless of the objective: even a perfect objective cannot produce a topology-faithful map from $\mathcal{Q}$ into a space too small to embed it. Graded compression destroys distinctions because the task rewards it; the floor entangles distinctions because the geometry requires it.

3.4 The floor is not a practical design target

The bound is loose in practice. Frankle and Carlin [2019] showed that winning lottery-ticket subnetworks train successfully only when reset to their original initialisation, suggesting that the structural information learned during training is entangled with specific weight values — it is not stored in the topology of the subgraph alone. A plausible implication is that networks may use surplus dimensions of Z to distribute and stabilise learned quotient structure across many weights. The dimension lower bound is therefore a *floor*, not a target: in practice, architectures may require substantially more capacity than $D_{\text{ambient}} \geq \text{emb}_{\text{top}}(\mathcal{Q})$ when the learned structure is distributed rather than cleanly localised.

Remark 3.8 (The bound is on the exposed representation, not on layer widths). The theorems above constrain the *latent ambient* Z — the representation produced by the encoder as a whole, before any prediction head. They say nothing about the dimension of any individual layer, intermediate activation, or architectural bottleneck inside the network. A deep network with a narrow internal layer can produce an exposed representation in $\mathbb{R}^m$ for any m; the theorems require that $m \geq \text{emb}_{\text{top}}(\mathcal{Q})$ but are silent on how the network achieves this internally. In particular, the theorems should *not* be read as implying that every layer must have width at least $\text{emb}_{\text{top}}(\mathcal{Q})$, or that a network with any layer of width less than $\text{emb}_{\text{top}}(\mathcal{Q})$ cannot produce a topology-faithful exposed representation.

3.5 Expressive power does not imply topological faithfulness

Remark 3.9 (Universal approximation does not bypass the floor). Universal approximation theorems [Cybenko, 1989, Hornik, 1991] guarantee that for any continuous target function, a sufficiently expressive network can approximate it in value. A network can be a universal function approximator and still induce a $\widetilde{E}$ that is non-injective or not a homeomorphism onto its image. Value universality and topology universality are independent properties.

3.6 Illustration: the phase cylinder

Return to the cylinder running example. $\mathcal{Q} = S^1$ requires $D_{\text{ambient}} \geq \text{emb}_{\text{top}}(S^1) = 2$. A scalar latent ($D_{\text{ambient}} = 1$) is below the floor: S^1 does not embed in $\mathbb{R}$. Whatever representation a scalar latent produces, it must collapse phases that the declared quotient keeps distinct. This is independent of the prediction target, the optimiser, and the training data.

4 Graded Compression

Once the floor is met, the criterion still permits latent shapes of intrinsic dimension strictly below $\text{emb}_{\text{top}}(\mathcal{Q})$ — shapes that approximate an embedding of a coarsening of $\mathcal{Q}$ rather than $\mathcal{Q}$ itself. This section establishes that this is not an artefact of optimisation or finite data: it is a structural feature of the criterion. Throughout this section, representations may take values in arbitrary measurable codomains, not necessarily $Z = \mathbb{R}^{D_{\text{ambient}}}$. The compression preorder is defined across codomains.

4.1 The population-level case

Definition 4.1 (Sufficient representation)**.** A representation E is *sufficient* for f if there exists a measurable map g such that $f = g \circ E$.

Definition 4.2 (Compression preorder)**.** For representations E_1 and E_2, write $E_1 \preceq_{\text{comp}} E_2$ if there exists a measurable map h such that $E_1 = h \circ E_2$. We say E_1 is *at least as compressed as* E_2.

Definition 4.3 (Minimal sufficient representation)**.** A representation E is *minimal sufficient* for f if it is sufficient for f and $E \preceq_{\text{comp}} E'$ for every sufficient E'.

Theorem 4.4 (Task-Map Collapse)**.** *Parts (i) and (ii) restate, in the compression-preorder framework, the classical result that the label function is the minimal sufficient statistic [Lehmann and Scheffé, 1950, Tishby et al., 1999]. Part (iii) connects injectivity of the task map to topological embeddability of the quotient.*

Define the label representation by $E_{\text{label}}(x) = f(x)$. Then:

- *(i) E_{label} is sufficient for f.*
- *(ii) E_{label} is the minimum of the compression preorder restricted to sufficient representations: $E_{\text{label}} \preceq_{\text{comp}} E$ for every sufficient E.*
- *(iii) Unless $t : \mathcal{Q} \to Y$ is injective, E_{label} does not embed $\mathcal{Q}$. Specifically, the map induced on the quotient is the task map t itself, and if t identifies two distinct quotient classes then E_{label} cannot be a topological embedding of $\mathcal{Q}$.*

Proof. See Appendix B. □

Remark 4.5 (Uniqueness of the minimum). The minimum element E_{label} is unique in the compression preorder among sufficient representations: any two representations that are each minimal must be equivalent up to relabelling, since each factors through the other. The minimum is the unique canonical element at the bottom of the sufficiency order, determined entirely by the task map f. A Kolmogorov complexity interpretation of why E_{label} is specifically the minimum is developed in Paper II [Farkas, 2026b, Remark 2.2a]: $E_{\text{label}} = t \circ q$ is the KC-simplest sufficient encoder, and simplicity bias in deep networks makes it the preferred solution under any simple task-loss objective.

The substantive content is part (iii). The image $t(\mathcal{Q})$ may have the same covering dimension as $\mathcal{Q}$ (sine example) but strictly lower *embedding* dimension when $\mathrm{emb}_{\mathrm{top}}(t(\mathcal{Q})) < \mathrm{emb}_{\mathrm{top}}(\mathcal{Q})$. Part (iii) says: the global minimum of the compression preorder among sufficient representations is structurally a coarsening of $\mathcal{Q}$. The criterion's bottom is, by construction, an embedding of $t(\mathcal{Q})$ rather than of $\mathcal{Q}$.

4.2 The finite-sample case: graded simplicial spans

The population-level case fixes one specific coarsening: the image $t(\mathcal{Q})$ under the task map. The finite-sample case admits every coarsening dimension from 1 up to the floor.

Theorem 4.6 (Finite-Sample Traversal; classical: Whyburn 1942, Rourke and Sanderson 1972)**.** *Let $D = \{x_1, \dots, x_n\} \subset \mathbb{R}^m$ be a finite set of pairwise distinct points. Then there exists a continuous injective map $\gamma : [0,1] \to \mathbb{R}^m$ and parameters $0 < t_1 < \cdots < t_n < 1$ such that $\gamma(t_i) = x_i$ for all i.*

Proof. Classical; piecewise-linear interpolation along a generic projection direction. See Appendix B. □

The traversal theorem is the $k = 1$ case of the more general graded construction below.

Theorem 4.7 (Graded Simplicial Spans)**.** *Let $D = \{x_1, \dots, x_n\}$ be in affine general position up to order $k+1$ in $\mathbb{R}^m$, and let $1 \le k \le \min(n-1, m)$. For every k-dimensional abstract simplicial complex K on n labelled vertices using each vertex at least once and using only $(k+1)$-tuples of affinely independent vertices, the linear-extension realisation map $|K| \to \mathbb{R}^m$ is a continuous map. Its image is a k-dimensional polyhedral subset of $\mathbb{R}^m$ containing D.*

Proof. Affine general position implies each chosen $(k+1)$-tuple is affinely independent. The image is a finite union of k-simplices, hence a k-dimensional polyhedral subset. □

Remark 4.8 (Geometric realisation). The image need not be a PL-embedded simplicial complex — different simplices may intersect in their interiors. The realisation is still a continuous map onto a k-dimensional polyhedral image; this is what the construction needs. The weakening from PL embedding to continuous polyhedral realisation is essential and is what makes the construction robust to the combinatorics of K.

Corollary 4.9 (Lower bound on simplicial-fit multiplicity)**.** *Let D be a finite set of size n in affine general position. For every k with $1 \le k \le \min(n-1, D_{\text{ambient}})$ the number $N_k(n)$ of distinct k-dimensional simplicial fits through D satisfies*

$$N_k(n) \;\ge\; \frac{n!}{2} \cdot \binom{n}{k+1}.$$

Proof. $n!/2$ undirected Hamiltonian paths through D. Each can be augmented with a k-simplex on $\binom{n}{k+1}$ choices of affinely independent vertices. □

Remark 4.10 (Looseness). The bound is loose; tighter combinatorial counts (e.g., k-skeleta of the $(n-1)$-simplex) grow much faster. The loose bound suffices for establishing factorial-times-polynomial growth at every k.

4.3 Sample-by-sample fit does not identify the manifold

Corollary 4.11 (Continuous representations through simplicial fits)**.** *For every k with $1 \leq k \leq \min(n-1, D_{\text{ambient}})$ and every k-dimensional simplicial fit through D, there exists a continuous representation $E : \mathcal{X} \to Z$ with latent shape of intrinsic dimension at most k (generically equal to k), achieving perfect training accuracy.*

Proof. Tietze extension applied to the simplicial fit, lifted to a continuous encoder. See Lemma 4.13 below. □

Remark 4.12 (Scope of the simplicial-span family). The simplicial-span construction is one particular family. Standard parametric interpolation provides additional families — for instance, smooth parametric interpolants exist for $m \geq k+1$ — and these are not counted in $N_k(n)$. The bound therefore counts only one structured family of fits; the true multiplicity of k-dimensional perfect-accuracy representations is much larger.

4.4 Continuous extension to the input space

Lemma 4.13 (Continuous extension; component-wise Tietze)**.** *Let $\mathcal{X}$ be a compact metric space and $D \subset \mathcal{X}$ a finite subset. Every map $\tau : D \to \mathbb{R}^d$ extends to a continuous map $\bar{\tau} : \mathcal{X} \to \mathbb{R}^d$ via the Tietze extension theorem applied component-wise. If $\mathcal{X}$ is connected, $\bar{\tau}(\mathcal{X})$ is connected.*

Proof. Each coordinate function $\tau^{(j)} : D \to \mathbb{R}$ is bounded and continuous on the closed subset D of the normal space $\mathcal{X}$. Tietze's theorem extends each $\tau^{(j)}$ continuously to $\mathcal{X}$. The vector-valued extension $\bar{\tau} = (\bar{\tau}^{(1)}, \ldots, \bar{\tau}^{(d)})$ is then continuous. The extension lands in $\mathbb{R}^d$, possibly slightly outside the simplicial complex $|K|$ but still on a k-dimensional set when $\bar{\tau}$ is constructed component-wise from the simplicial fit. Connectedness follows from continuity of $\bar{\tau}$ and connectedness of $\mathcal{X}$. □

Note on extension targets. The lemma uses no ANR retraction hypothesis. The extension lands in $\mathbb{R}^d$, not necessarily in $|K|$ itself; the statement is about the existence of a continuous map agreeing with τ on D, not about preservation of polyhedral structure off the data. This weakening is what makes the lemma applicable to general compact metric $\mathcal{X}$ without further assumptions on the simplicial fit.

4.5 Illustration: graded compression on the cylinder

For the cylinder running example, $\mathcal{Q} = S^1$, $\mathcal{X} = S^1 \times [0,1]$, target $t(\theta) = \sin\theta$, latent ambient $Z = \mathbb{R}^{D_{\text{ambient}}}$ with $D_{\text{ambient}} \geq 2$.

Population-level case. The label representation $E_{\text{label}}(x) = \sin(\theta(x))$ is the minimum of the compression preorder. It is sufficient. Its image is the interval $[-1,1]$, which has covering dimension 1 (same as S^1) but embedding dimension 1 (strictly less than $\text{emb}_{\text{top}}(S^1) = 2$). The induced map on $\mathcal{Q}$ is the task map t itself, which identifies θ with $\pi - \theta$. The latent shape is an interval, not a circle.

Finite-sample case. For a sample of n points in the cylinder, the criterion admits at least $\frac{n!}{2}$ one-dimensional traversal fits (Theorem 4.6, $k = 1$); at least $\frac{n!}{2}\binom{n}{2}$ two-dimensional simplicial fits (the $k = 2$ case of Theorem 4.7); and continues up to the floor at $k = 2$ for the cylinder's full embedding. Each fit is a continuous representation (Corollary 4.11) achieving perfect training accuracy.

What the criterion sees. In both cases, the criterion sees the prediction-target accuracy and nothing else. The traversal representation has no concept of the wrap-around at $\theta = 2\pi \equiv 0$. The label representation has flattened S^1 to an interval. The criterion does not distinguish either from a faithful S^1-embedding; among them, simplicity bias selects the lowest-dimensional fit, which is the most coarsened.

4.6 Forward pointer to the series

The combinatorial structure at each k — which k-dimensional fits a given architecture realises, and how the optimiser navigates among them — is the subject of Paper II [Farkas, 2026b], where stochastic escape mechanisms and the graded-compression structure are developed. The present paper establishes the structural fact that the criterion permits the entire graded family; the dynamical question of which member is selected by SGD is a separate concern.

5 Quotient-Respect Failure

§§3 and 4 analysed structural failures of representations that *do* respect the declared quotient. §3 showed that, even with quotient respect, the latent ambient may be too small to embed $\mathcal{Q}$. §4 showed that, with both quotient respect and an adequate latent ambient, the criterion may select an encoder approximating an embedding of a coarsening of $\mathcal{Q}$ rather than $\mathcal{Q}$ itself. Both analyses presuppose that an induced map $\widetilde{E} : \mathcal{Q} \to Z$ exists; they study its properties.

This section asks what happens when the encoder does not respect the declared quotient: when there exist $\sim$-equivalent observations that the encoder distinguishes.

5.1 Quotient-respect failure as a structural failure

Define the *encoder-induced relation* $\sim_E$ on $\mathcal{X}$ by

$$x \sim_E x' \iff E(x) = E(x'),$$

and write $Q_E := \mathcal{X}/\sim_E$. Quotient respect (Definition 2.5) is the relation-theoretic statement $\sim \subseteq \sim_E$: every $\sim$-equivalent pair is $\sim_E$-equivalent.

Definition 5.1 (Quotient-respect failure)**.** An encoder $E : \mathcal{X} \to Z$ exhibits *quotient-respect failure* with respect to q when $\sim \not\subseteq \sim_E$: there exist $x, x' \in \mathcal{X}$ with $x \sim x'$ and $E(x) \neq E(x')$.

The condition is asymmetric. It says the encoder fails to identify some pairs that the declared quotient identifies; it does not say anything about whether the encoder makes additional identifications of its own. This asymmetry produces two structurally distinct subcases.

Pure refinement (nuisance retention). $\sim_E \subsetneq \sim$. The encoder distinguishes some $\sim$-equivalent points but never collapses points the declared quotient keeps distinct. Q_E is a refinement of $\mathcal{Q}$ — it sits between $\mathcal{X}$ and $\mathcal{Q}$ on the lattice of equivalence relations on $\mathcal{X}$, and there is a canonical surjection $Q_E \twoheadrightarrow \mathcal{Q}$.

Incomparable failure. $\sim_E$ and $\sim$ are incomparable: $\sim_E \not\subseteq \sim$ and $\sim \not\subseteq \sim_E$. The encoder both distinguishes some $\sim$-equivalent points *and* identifies some points the declared quotient keeps distinct. Q_E is not a refinement of $\mathcal{Q}$, nor a coarsening; it is a different quotient of $\mathcal{X}$, related to $\mathcal{Q}$ in a more complex way.

The remaining algebraic case, $\sim \subsetneq \sim_E$, is excluded by definition: it would require the encoder to identify everything $\sim$ identifies and more, which is precisely the regime of §4 (the encoder respects q and coarsens $\mathcal{Q}$ further), not §5.

When quotient-respect fails — in either subcase — no induced map $\widetilde{E} : \mathcal{Q} \to Z$ exists, and the analyses of §§3 and 4 are precluded as stated. The structural relationship the paper is about — between $\mathcal{Q}$ and the latent shape $E(\mathcal{X})$ — cannot be expressed at the level of $\mathcal{Q}$, because the latent shape has no canonical map *from* $\mathcal{Q}$. This is a hard boundary, not a matter of degree.

The two subcases are not equally common, and they admit different remedies. The pure-refinement subcase is the standard nuisance-retention failure most readers will picture: the encoder "remembers too much." The incomparable subcase is the more dangerous one for diagnosis, because the encoder both forgets what it should remember and remembers what it should forget — and either symptom alone can mask the other when checked in isolation.

5.2 What the criterion does not enforce

Standard training criteria do not, by themselves, enforce quotient respect. Three observations make this concrete.

Bare prediction loss is silent. The label representation $E_{\text{label}}(x) = f(x) = t(q(x))$ trivially factors through q and achieves zero prediction loss. So does any encoder E that retains q-information *plus arbitrary nuisance*: as long as some downstream map $g : Z \to Y$ recovers f from E, the prediction loss is zero whether or not E factored through q. The criterion measures the relationship between E and the prediction target; it does not interrogate the relationship between E and q.

Sufficiency does not enforce respect. A representation E is *task-sufficient* when there exists $g : Z \to Y$ with $f = g \circ E$. Sufficiency is a property of the relationship between E and f; it does not constrain how E treats nuisance variation. A sufficient E can distinguish $x \sim x'$ as long as g subsequently identifies them, and the sufficiency criterion is satisfied.

Information-bottleneck objectives do not enforce respect either. Bottleneck criteria penalise the mutual information $I(E; \mathcal{X})$ subject to preserving $I(E; Y)$. A representation that retains nuisance can still be a bottleneck minimiser when the nuisance correlates with the prediction target — for instance, when nuisance carries instance-level identifiers that are spuriously predictive, or when the bottleneck temperature is finite and the nuisance carries little but nonzero information about Y. The bottleneck objective targets compression of $\mathcal{X}$, not factorisation through q.

Each of these observations covers both subcases. The criterion's silence on quotient respect does not depend on whether the failure is pure refinement or incomparable; it follows from the criterion's structural blindness to the relationship between E and q.

5.3 The embedding-space description

The embedding-space lens of §1.3 places quotient-respect failure as follows. The encoder E factors through $q_E : \mathcal{X} \to Q_E$ by construction, and the induced map $Q_E \to Z$ is injective. The encoder respects *some* quotient — namely $\sim_E$ — but not necessarily the declared one. If the latent shape $E(\mathcal{X})$ is a topological embedding of Q_E — if there is a homeomorphism between Q_E and $E(\mathcal{X})$ — then the encoder is *faithful*, but to Q_E rather than to $\mathcal{Q}$. The encoder approximates an embedding $\phi' : Q_E \hookrightarrow \mathbb{R}^{D_{\text{ambient}}}$ from a quotient of $\mathcal{X}$ different from $\mathcal{Q}$.

The structural relationship between Q_E and $\mathcal{Q}$ depends on the subcase.

Pure refinement. When $\sim_E \subsetneq \sim$, the canonical surjection $Q_E \twoheadrightarrow \mathcal{Q}$ exists. The encoder is faithful to a refinement: it has retained nuisance variation as latent coordinates without compromising any $\sim$-equivalence the declared quotient asserts. Two extremes mark the spectrum: *no identification*, where $\sim_E$ is the identity on $\mathcal{X}$ and $Q_E = \mathcal{X}$ (the encoder injects $\mathcal{X}$ into Z and treats every observation as distinct); and *maximal compatible identification*, where $\sim_E$ is the finest equivalence relation strictly finer than $\sim$ that the encoder happens to realise. In between, Q_E may pick out specific nuisance variations the encoder retains: amplitude on the phase cylinder, pose in image classification, instance identity in metric learning, speaker identity in speech models.

Incomparable failure. When $\sim_E$ and $\sim$ are incomparable, no surjection between Q_E and $\mathcal{Q}$ exists in either direction. There is a canonical *common refinement* — the equivalence generated by $\sim_E \cup \sim$ — and a canonical *common coarsening*, but Q_E itself does not sit on the lattice path between $\mathcal{X}$ and $\mathcal{Q}$. The encoder is faithful to a quotient that is structurally different from $\mathcal{Q}$: it identifies along axes the declared quotient does not, and distinguishes along axes the declared quotient identifies. This subcase protects the section from an overstatement of the form "quotient-respect failure means the encoder is faithful to a more detailed version of $\mathcal{Q}$." It does not, in general.

The embedding-space description does not *define* quotient-respect failure — the structural definition is the failure of $\sim$ to be contained in $\sim_E$. It places and understands the failure: by characterising Q_E and its relationship to $\mathcal{Q}$, it tells us not just that the encoder turned out faithful to a different quotient, but to which one.

5.4 Sources of quotient-respect failure

Quotient-respect failure does not arise from a single mechanism. Several distinct sources produce it, and they tend to produce different subcases.

Architectural insensitivity. When the architecture has no commitment to $\sim$ — when it is a generic MLP or transformer with no built-in invariances — the encoder is free to retain features that improve training accuracy, including nuisance features that happen to correlate with the prediction target. This source typically produces the pure-refinement subcase: the encoder retains more than q requires but does not actively collapse along directions q keeps distinct.

Augmentation mismatch. Self-supervised and contrastive methods enforce equivalence through augmentation: they declare that a sample and its augmentation should map to similar latent points. The augmentation distribution defines an equivalence relation $\sim_{\text{aug}}$, and the encoder is shaped to respect $\sim_{\text{aug}}$ rather than $\sim$. When $\sim_{\text{aug}}$ is *strictly finer* than $\sim$ — when augmentation under-identifies relative to the declared quotient — the encoder respects a refinement of $\mathcal{Q}$, producing the pure-refinement subcase. When $\sim_{\text{aug}}$ is *strictly coarser*, the encoder respects a coarsening; this is a graded-compression phenomenon under a different equivalence relation, not a respect failure. When $\sim_{\text{aug}}$ is *incomparable* with $\sim$ — for instance, when the augmentation distribution mixes lawful nuisance identifications with extraneous identifications across distinct $\sim$-classes — the encoder produces the incomparable subcase.

Symmetry mismatch. When the architecture has built-in invariances under a group G_{arch} that is not contained in (and does not contain) the symmetry group G generating $\sim$, the encoder respects G_{arch}-orbits rather than G-orbits. Periodicity miscalibration is a paradigm case: an encoder using $\cos 2\theta$ when the true period is 2π is invariant under a half-period shift the data are not, and the resulting failure is incomparable.

Implicit nuisance signal. When training data carry instance-level identifiers that are individually predictive — speaker embeddings, image patches with serial-number-like artefacts, sequence positions in time series, dataset-specific watermarks — the encoder may incorporate these into its representation without violating any criterion. The result is a Q_E much finer than $\mathcal{Q}$ — pure refinement.

Overparameterisation. In overparameterised regimes the encoder has spare capacity to retain features the criterion does not require. Whether overparameterisation amplifies or dampens quotient-respect failure, and which subcase it favours, depends on training dynamics and inductive bias of the optimiser, which lie outside the scope of this paper. Paper II [Farkas, 2026b] addresses this side of the picture as part of its admissibility-source taxonomy.

The list is not exhaustive. It indicates that quotient-respect failure is a family of structurally similar failures with different practical signatures and different routes to remedy.

5.5 Illustration: nuisance retention and periodicity miscalibration on the cylinder

The cylinder running example separates the failure modes cleanly. The observation space is $\mathcal{X} = S^1 \times [0,1]$, the lawful quotient is $q : \mathcal{X} \to S^1$ projecting to phase, and the prediction target is $t(\theta) = \sin\theta$. A faithful encoder respects q: it maps points of equal phase to equal latent points, regardless of amplitude, and its induced map $\widetilde{E} : S^1 \to Z$ is a topological embedding of the circle.

Pure-refinement example. Consider the encoder $E_1 : \mathcal{X} \to \mathbb{R}^3$ defined by

$$E_1(\theta, r) = (\cos\theta, \sin\theta, r).$$

This E_1 achieves zero prediction loss: $\sin\theta$ is recovered from the second coordinate via $g(z_1, z_2, z_3) = z_2$. It is task-sufficient. It is also a topological embedding of $\mathcal{X}$ into $\mathbb{R}^3$ — the latent shape is the cylinder itself, embedded as a surface. But E_1 does not factor through q: distinct amplitudes at the same phase map to distinct latent points. In the relation-theoretic language, $\sim_{E_1}$ is the identity on $\mathcal{X}$, and $\sim_{E_1} \subsetneq \sim$. In the embedding-space language, E_1 approximates the embedding $\phi' : \mathcal{X} \hookrightarrow \mathbb{R}^3$ given by the same formula, where ϕ' is an embedding of the *unquotiented* observation space. The encoder is faithful to $\mathcal{X}$, not to S^1. This is the pure-refinement subcase.

Incomparable example. Consider the encoder $E_2 : \mathcal{X} \to \mathbb{R}^3$ defined by

$$E_2(\theta, r) = (\cos 2\theta, \sin 2\theta, r).$$

This E_2 also fails quotient respect, but in a structurally different way. Distinct amplitudes at the same phase still map to distinct latent points, so $\sim \not\subseteq \sim_{E_2}$. But additionally, the encoder identifies θ with $\theta + \pi$: phases the declared quotient q keeps distinct are mapped to the same latent point at fixed r. So $\sim_{E_2} \not\subseteq \sim$ as well. The two relations are incomparable.

The encoder E_2 is faithful to the quotient $\mathcal{X}/\sim_{E_2}$, where $\sim_{E_2}$ identifies (θ, r) with $(\theta + \pi, r)$. This quotient is structurally different from S^1: it is a half-period quotient of the phase combined with full retention of amplitude. The latent shape is a smaller cylinder, traversed twice as the phase runs from 0 to 2π.

What makes the incomparable case dangerous is that E_2 may still achieve respectable prediction loss when the prediction target is $t(\theta) = \sin\theta$. The half-period collapse of θ and $\theta + \pi$ does not align with the declared task structure (since $\sin\theta \neq \sin(\theta + \pi)$ in general), so the criterion does not certify E_2 as task-sufficient under this t. But under a different prediction target — for example $t(\theta) = \sin 2\theta$ — E_2 becomes task-sufficient and the incomparable failure is fully concealed by the criterion. The pattern generalises: whenever the prediction target is invariant under a non-lawful symmetry the encoder happens to realise, the incomparable subcase is invisible to bare loss.

Bottleneck remediation does not, in general, fix either subcase. If amplitude variation in the data is large compared to the bottleneck temperature, or if amplitude correlates spuriously with the prediction target through finite-sample artefacts, the bottleneck minimiser retains r. If the prediction target shares a non-lawful symmetry with the architecture, the bottleneck minimiser may align with that symmetry rather than with $\sim$. The criterion is satisfied throughout; the structural failure is invisible to it.

Detection of either subcase requires either an explicit declaration of $\sim$ — for instance, an equivariance loss penalising $E(\theta, r) \neq E(\theta, r')$ — or an architectural commitment that builds the equivalence in (e.g., max-pooling over the amplitude axis, or a group-equivariant layer with the cylinder's $\mathbb{R}_{>0}$-action). Distinguishing pure refinement from incomparable failure once detection has occurred requires a separate test: probing whether the encoder identifies pairs that should be distinct, not just whether it distinguishes pairs that should be identified.

5.6 Dimensional consequences

When E approximates an embedding of Q_E, the latent ambient must accommodate Q_E, not $\mathcal{Q}$. Dimensional adequacy in this regime is

$$D_{\text{ambient}} \geq \text{emb}_{\text{top}}(Q_E),$$

which can be substantially larger than the floor $\text{emb}_{\text{top}}(\mathcal{Q})$ developed in §3. For the pure-refinement example above, $\text{emb}_{\text{top}}(S^1) = 2$ but $\text{emb}_{\text{top}}(S^1 \times [0,1]) = 3$. For the incomparable example, $\text{emb}_{\text{top}}(Q_E) = 3$ as well. In general, the dimensional demand of the incomparable subcase need not exceed the pure-refinement subcase, but it is also not bounded below by it: an incomparable Q_E is incomparable along the lattice, not along the dimensional ladder.

This is a downstream remark, not the main claim of the section. The structural failure is the loss of factorisation through q, not the dimensional consequence of choosing to be faithful to a different quotient. A practical corollary is worth recording, however: under quotient-respect failure, *increasing* the latent ambient does not in general reduce the failure. More room makes more Q_E-like targets available, not fewer. The remedy is to constrain the encoder to factor through q, by architectural commitment or by an explicit quotient-respect penalty — not to expand the latent ambient.

5.7 Locatello unidentifiability as the parallel manifestation

The unidentifiability theorem of Locatello et al. [2019] establishes that, in the absence of an external admissibility declaration, no purely loss-based procedure can identify the correct disentangled factors of variation. This is the parallel manifestation of criterion silence in the regime where $\sim$ has not been declared. The three failure modes of this paper are the parallel manifestations in the regime where $\sim$ has been declared. They are not downstream of the Locatello result; they are facets of the same underlying problem.

Together they describe the criterion's silence on the entire structural relationship between the encoder and the generating process: silence on $\sim$ when it is unknown (Locatello), silence on the architectural floor for $\text{emb}_{\text{top}}(\mathcal{Q})$ once $\sim$ is known (§3), silence on whether the encoder approximates an embedding of $\mathcal{Q}$ rather than a coarsening (§4), and silence on whether the encoder respects the declared equivalence at all (§5). The embedding-space program of §1.3 is the structural account that would let faithfulness be certified if all these structural inputs were supplied; the criterion supplies none of them.

5.8 Summary and bridge to §6

The three failure modes — embedding floor (§3), graded compression (§4), and quotient-respect failure (§5) — are logically independent. §§3 and 4 presuppose quotient respect; §5 analyses what happens when that presupposition fails. None of the three implies the other two: an encoder may satisfy quotient respect and still fail the floor; it may satisfy the floor and respect q and still pick a coarsening of $\mathcal{Q}$; it may satisfy the floor and pick a different quotient Q_E — refinement or incomparable — without ever factoring through q.

Within §5, the asymmetry of the quotient-respect condition produces the two subcases recorded in §5.1: pure refinement, in which the encoder is faithful to a more detailed quotient than $\mathcal{Q}$, and incomparable failure, in which the encoder is faithful to a structurally different quotient. The two subcases share the same root failure — $\sim \not\subseteq \sim_E$ — and admit similar architectural and loss-based remedies, but they call for different diagnostics.

The next section assembles the three failure modes into the structural account of lawful abstraction.

6 Lawful Abstraction

6.1 The three requirements

Proposition 6.1 (Lawful Abstraction Decomposition). *For a representation $E : \mathcal{X} \to Z$, the following three properties are logically distinct. No one of them implies the conjunction of the other two.*

(I) Task sufficiency. *There exists a measurable map $g : Z \to Y$ such that $f = g \circ E$.*
(II) Quotient respect. *There exists a measurable map $\widetilde{E} : \mathcal{Q} \to Z$ such that $E = \widetilde{E} \circ q$.*
(III) Structural faithfulness. *The induced map $\widetilde{E} : \mathcal{Q} \to Z$ is a topological embedding of $\mathcal{Q}$. In stronger settings, $\widetilde{E}$ is additionally required to be bi-Lipschitz.*

A lawful abstraction *is a representation satisfying all three.*

The three properties are logically independent: no two imply the third. The label representation $E_{\text{label}} = t \circ q$ satisfies (I) and (II) while failing (III) whenever t is non-injective. A topological embedding $\widetilde{E}$ composed with q satisfies (II) and (III) while potentially failing (I). A sufficient representation that retains nuisance alongside predictive features satisfies (I) while failing (II).

The three failure modes of §§3, 4, 5 map onto the three conditions:

- *Embedding floor (§3):* condition (III) cannot be satisfied at all, because the latent ambient Z does not host any homeomorphic image of $\mathcal{Q}$.
- *Graded compression (§4):* condition (II) is satisfied — the encoder factors through q — but condition (III) generically fails within (II), and the criterion is not certified to detect this. The induced map $\widetilde{E} : \mathcal{Q} \to Z$ is non-injective, except in special cases where the ordering of sufficient representations aligns with the quotient structure (i.e., when the task map t is itself injective).
- *Quotient-respect failure (§5):* condition (II) fails directly. Condition (III) is then unstatable at the level of $\mathcal{Q}$, since no induced map $\widetilde{E} : \mathcal{Q} \to Z$ exists.

6.2 The embedding-space lens on the decomposition

The three conditions can be read off the embedding-space program of §1.3 as separate aspects of the relationship between $\mathcal{Q}$, the encoder's chosen embedding ϕ, and the latent shape $E(\mathcal{X})$.

Task sufficiency (I) is the relationship between E and the prediction target f. It is silent on the embedding-space program: E may be sufficient for f regardless of which ϕ the encoder approximates and from which source quotient. Among the three failure modes, only graded compression interacts with sufficiency directly — and it does so by selecting an embedding from a coarsening of $\mathcal{Q}$ that the criterion treats as equivalent to a $\mathcal{Q}$-embedding for the purposes of predicting f.

Quotient respect (II) is the relationship between E and the declared quotient q. In embedding-space language: it says the encoder approximates an embedding from $\mathcal{Q}$, not from a different quotient of $\mathcal{X}$. Quotient-respect failure (§5) is the structural negation;

the encoder approximates an embedding from $\mathcal{X}/\sim_E$ where $\sim_E$ does not contain $\sim$, and the source quotient is either a refinement of $\mathcal{Q}$ (pure-refinement subcase) or incomparable with $\mathcal{Q}$ (incomparable subcase).

Structural faithfulness (III) is the relationship between the induced map $\widetilde{E}$ and the topology of $\mathcal{Q}$. It says the embedding from $\mathcal{Q}$ that the encoder approximates is a *topological* embedding rather than a continuous-but-collapsing map. The embedding floor (§3) is the architectural condition that an embedding from $\mathcal{Q}$ exists at all in the latent ambient; graded compression (§4) is the failure of (III) when the floor is met but the criterion selects an embedding of a coarsening instead.

The three conditions are logically independent, and the three failure modes are correspondingly independent: condition (I) can hold while either (II) or (III) fails; (II) can hold while (III) fails (graded compression) or while (I) fails (a quotient-respecting non-sufficient encoder); (III) can hold while (II) fails (an encoder that topologically embeds some quotient of $\mathcal{X}$ different from $\mathcal{Q}$). The three failure modes correspond to disjoint structural defects of the embedding-space relationship, and standard training criteria certify none of them.

6.3 Relation to disentanglement

The central result of Locatello et al. [2019] is that unsupervised disentanglement is impossible without inductive biases: the correct factorisation cannot be identified from data alone. In the framework of this paper, the Locatello result is the parallel manifestation of criterion silence in the regime where $\sim$ has not been externally declared: the criterion does not allow $\sim$ itself to be identified from data alone. The three failure modes of this paper are the parallel manifestations in the regime where $\sim$ has been declared. They are not downstream of the Locatello result — they are facets of the same underlying problem of criterion silence.

Together, they describe the criterion's silence on the entire structural relationship between the encoder and the generating process: silence on $\sim$ when it is unknown (Locatello), silence on the architectural floor once $\sim$ is known (§3), silence on whether the encoder approximates an embedding of $\mathcal{Q}$ rather than a coarsening (§4), and silence on whether the encoder respects the declared equivalence at all (§5). The embedding-space program of §1.3 is the structural account that would let faithfulness be certified if all these structural inputs were supplied; the criterion supplies none of them.

7 Metric Distortion: A Refinement

Topology-faithfulness is necessary but not sufficient for downstream reasoning. A homeomorphism can distort distances arbitrarily while remaining valid throughout. This section identifies metric distortion as a refinement of the topology-faithful condition: even when all three architectural conditions are met and the encoder respects the quotient, the criterion does not certify that the embedding's geometric quality is preserved.

7.1 The topology-geometry gap

Proposition 7.1 (Topology–Geometry Gap; cf. Burago et al. 2001)**.** *Let* $\mathcal{Q} = [0, 1]$ *with the standard metric. For* $\alpha \geq 1$ *define* $\widetilde{E}_\alpha(x) = x^\alpha$*. Each* $\widetilde{E}_\alpha$ *is a homeomorphism of* $[0, 1]$

onto itself, hence topology-faithful. However, the metric distortion of $\widetilde{E}_\alpha$ is unbounded: as $\alpha \to \infty$, neighbourhoods of fixed size near 0 collapse to arbitrarily small images.

Proof. See Appendix B. □

Nearest-neighbour retrieval, clustering, interpolation between latent codes, and any other operation that treats distances in Z as proxies for semantic similarity all depend on the metric structure of Z approximating the metric structure of $\mathcal{Q}$. A topology-faithful encoder that achieves perfect prediction loss can fail all of these.

7.2 Metric-faithful representations

Definition 7.2 (Metric-faithful representation)**.** A continuous map $\widetilde{E} : \mathcal{Q} \to \mathbb{R}^m$ is *metric-faithful* if it is bi-Lipschitz: there exist constants $0 < c \leq C < \infty$ such that

$$c \cdot d_{\mathcal{Q}}(q_1, q_2) \leq \|\widetilde{E}(q_1) - \widetilde{E}(q_2)\| \leq C \cdot d_{\mathcal{Q}}(q_1, q_2)$$

for all $q_1, q_2 \in \mathcal{Q}$.

Remark 7.3 (The hierarchy)*.* The structural requirements on $\widetilde{E}$ form a strict hierarchy:

$$\text{metric-faithful} \implies \text{topology-faithful} \implies \text{continuous} \implies \text{measurable.}$$

Each arrow is strict. For lawful abstraction (Proposition 6.1), the minimum required is topology-faithful. For operations that rely on distance-based reasoning in Z, metric-faithful is required.

7.3 Where metric distortion arises in practice

Wang and Isola [2020] show that the contrastive loss asymptotically optimises alignment — collapsing nuisance-equivalent observations to nearby codes — and uniformity — preventing concentration of large regions of $\mathcal{Q}$ onto small regions of Z. In the language of this paper, uniformity is a soft form of the lower Lipschitz condition. Hofer et al. [2017] introduced differentiable topology layers that penalise deviations from a target topological structure, and Moor et al. [2023] proposed the Representation Topology Divergence (RTD). Both methods directly operationalise Definition 3.1; neither targets metric distortion.

7.4 Illustration: metric distortion on the phase cylinder

Suppose the learner applies a topology-enforcing objective that detects the missing one-cycle and closes the representation into a loop. Gradient descent closes the loop by collapsing consecutive turns at phases θ and $\theta + 2\pi/N$, producing a closed curve that is a topological embedding of S^1. The topology loss reports success.

The metric geometry is wrong. The bi-Lipschitz lower constant $c \to 0$ as $N \to \infty$. The topology loss has done exactly what it was asked to do. What it was not asked to do — and what no standard training objective asks for — is to distribute arc length faithfully. This is a fourth manifestation of criterion silence: even when topology-faithfulness has been achieved, metric quality is not certified.

8 Summary

The master claim of the paper is structural: predictive loss does not certify quotient faithfulness. The criterion measures the relationship between an encoder and a prediction target; faithfulness is a relationship between an encoder and the structure of the generating process. These are different relationships, and the predictive criterion is silent on the second.

The silence has two regimes. Without an external admissibility declaration, the criterion does not even allow faithfulness to be stated as a mathematical condition: Locatello et al. [2019] establish this for unsupervised disentanglement, and the same pattern recurs for any structural identification from loss alone. With an external admissibility declaration in hand, the criterion still admits encoders that fail every structural requirement faithfulness imposes. The three failure modes of this paper — embedding floor (§3), graded compression (§4), and quotient-respect failure (§5) — are concrete manifestations of criterion silence in the $\sim$-declared regime. They are the strongest form of the claim: even when the practitioner has the structural input, an unconstrained architecture, and an ideal optimiser, the criterion still does not certify faithfulness. Where the practitioner has less, the silence is broader still.

The embedding floor (§3). The condition $D_{\text{ambient}} \geq \text{emb}_{\text{top}}(\mathcal{Q})$ that the latent ambient must satisfy for any homeomorphic image of $\mathcal{Q}$ to exist inside it. Below the floor, no $\phi : \mathcal{Q} \hookrightarrow \mathbb{R}^{D_{\text{ambient}}}$ exists; the program cannot begin. The contribution is the placement of this condition as architectural adequacy, logically prior to any training-time concern.

Graded compression (§4). Above the floor, within the class of encoders that respect q, the criterion permits encoders approximating an embedding from a *coarsening* of $\mathcal{Q}$ rather than from $\mathcal{Q}$ itself. At the population level, when the prediction target t is non-injective on $\mathcal{Q}$, the global minimum of the deterministic compression preorder is the label representation $E_{\text{label}} = t \circ q$, with hosting requirement $\text{emb}_{\text{top}}(t(\mathcal{Q}))$ — strictly below $\text{emb}_{\text{top}}(\mathcal{Q})$ in the generic non-injective case. At the finite-sample level, any finite labelled sample admits a graded family of k-dimensional simplicial fits at every k from 1 up to the floor, with at least $\frac{n!}{2}\binom{n}{k+1}$ distinct continuous polyhedral realisations achieving perfect training accuracy.

Quotient-respect failure (§5). The encoder approximates an embedding from a quotient of $\mathcal{X}$ that does not factor through q at all. Two structurally distinct subcases arise. In the *pure-refinement* subcase, the encoder retains nuisance variation as a latent coordinate; the embedding ϕ' is from $\mathcal{X}/\sim_E$ for some $\sim_E$ strictly finer than $\sim$. In the *incomparable* subcase, the encoder both retains nuisance distinctions and collapses lawful ones — the encoder-induced relation $\sim_E$ is incomparable with $\sim$, and the resulting source quotient is structurally different from $\mathcal{Q}$. Both subcases share the same root failure ($\sim \not\subseteq \sim_E$); both are invisible to bare loss, sufficiency, and information-bottleneck objectives.

Faithful to the wrong source. In each failure mode the encoder is either prevented from being faithful to anything (§3) or is faithful to the wrong source (§§4 and 5). Faithfulness to $\mathcal{Q}$ requires the encoder to approximate an embedding from $\mathcal{Q}$ itself, neither coarsened, nor refined, nor replaced by a structurally different quotient of $\mathcal{X}$. The criterion does not enforce this; the program of §1.3 organises the architectural and training choices that ensure the right source is targeted when the criterion is the only training signal available.

Lawful abstraction. §6 makes the structural decomposition explicit: a representation is lawfully abstract when it is task-sufficient (I), quotient-respecting (II), and structurally faithful (III). Each condition must be stated and certified independently because no two imply the third. §7 identifies metric distortion as a fourth, refining condition: even when all three are met, the geometric quality of the embedding may be arbitrarily distorted by topology-faithful but bi-Lipschitz-violating maps.

Contribution. The paper's contribution is conceptual and organisational: the placement of the embedding-space program as the structural insight mediating between the declared quotient and the latent ambient; the placement of the embedding floor as architectural adequacy; the unification of population-level compression and finite-sample combinatorial underdetermination as cases of one graded phenomenon (Theorems 4.4 and 4.7); the identification of quotient-respect failure as the third manifestation, with refinement and incomparable subcases distinguished structurally (Definition 5.1); and the formalisations (compression preorder, quotient faithfulness criterion) that give precise language for the gap. The technical hinges are Theorem 4.4(iii), Theorem 4.7 with Corollary 4.9, and the asymmetry of the quotient-respect condition.

What the paper does not claim. The paper does not claim that standard training never produces faithful representations. Its narrower conclusion is that task performance alone does not certify faithfulness. When recovery happens, it cannot be credited to samplewise fit, sufficiency, or compression alone — some additional structure must be doing the work, for example architectural priors, augmentation-induced equivalences, stochastic optimisation bias, neighbourhood objectives, or other implicit admissibility signals. The present paper identifies the formal gap; the constructive question of how practical systems sometimes bridge it is deferred to the rest of the series.

Forward to the series. Paper II [Farkas, 2026b] addresses the recovery side: the admissibility sources, the role of stochastic escape, and the architectural conditions under which the failure modes are avoided. Paper III [Farkas, 2026c] addresses the architectural upper-bound question: which networks of which width and depth can realise faithful encodings of process-generated quotients. Paper V [Farkas, 2026e] maps the framework against existing traditions in representation learning theory. Paper VI [Farkas, 2026f] introduces generator-constrained discovery for the regime where the symmetry algebra of $\sim$ is partially specified. Paper VII [Farkas, 2026g] provides empirical validation on dynamical systems including the phase cylinder, the Hopf fibration, and the double pendulum. Paper IV generalises the framework from a single quotient to a quotient atlas, where $\mathcal{X}$ decomposes into regions with distinct local quotient structure.

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A Definitions and Key Concepts

This appendix collects the paper's formal definitions for quick reference.

Admissibility and quotient structure.

Admissible nuisance relation (Definition 2.1). An equivalence relation $\sim$ on $\mathcal{X}$ identifying observations that differ only in semantically irrelevant ways.

Lawful quotient (Definition 2.2). $(\mathcal{Q}, q)$ where $\mathcal{Q} = \mathcal{X}/{\sim}$ and $q : \mathcal{X} \twoheadrightarrow \mathcal{Q}$ is the canonical projection. $\mathcal{Q}$ is the space of semantic states.

Representation map / encoder (Definition 2.4). A measurable map $E : \mathcal{X} \to Z$ into the latent ambient $Z = \mathbb{R}^{D_{\text{ambient}}}$.

Quotient-respecting representation (Definition 2.5). E factors as $E = \widetilde{E} \circ q$; equivalently, $E(x) = E(x')$ whenever $x \sim x'$.

Encoder-induced relation (§5.1). $x \sim_E x' \iff E(x) = E(x')$. The encoder respects the quotient if and only if $\sim \subseteq \sim_E$.

Quotient-respect failure (Definition 5.1). $\sim \not\subseteq \sim_E$. Two subcases: pure refinement ($\sim_E \subsetneq \sim$) and incomparable failure ($\sim_E$ and $\sim$ neither containing the other).

Compression and sufficiency.

Sufficient representation (Definition 4.1). E is sufficient for f if $f = g \circ E$ for some measurable g.

Compression preorder (Definition 4.2). $E_1 \preceq_{\text{comp}} E_2$ iff $E_1 = h \circ E_2$ for some measurable h.

Minimal sufficient representation (Definition 4.3). E is sufficient and $E \preceq_{\text{comp}} E'$ for every sufficient E'.

Label representation. $E_{\text{label}}(x) = f(x)$. The unique $\preceq_{\text{comp}}$-minimum among sufficient representations (Theorem 4.4); collapses $\mathcal{Q}$ whenever t is non-injective.

Faithfulness conditions.

Topology-faithful representation (Definition 3.1). $\widetilde{E} : \mathcal{Q} \to Z$ is a topological embedding: a homeomorphism from $\mathcal{Q}$ onto $\widetilde{E}(\mathcal{Q}) \subset Z$.

Metric-faithful representation (Definition 7.2). $\widetilde{E}$ is bi-Lipschitz with constants $0 < c \leq C < \infty$.

Lawful abstraction (Proposition 6.1). A representation satisfying task sufficiency, quotient respect, and structural faithfulness simultaneously.

Architectural conditions and dimensional vocabulary.

Dimensional adequacy. $D_{\text{ambient}} \geq \text{emb}_{\text{top}}(\mathcal{Q})$.

Expressive adequacy. $\mathcal{H}$ contains a function whose induced map on $\mathcal{Q}$ is a topological embedding.

Realisation adequacy. The parameter configuration realises such a function accessibly under the chosen training procedure.

Embedding space. A pure mathematical object $\mathbb{R}^m \supset \phi(\mathcal{Q})$ for some chosen homeomorphism $\phi : \mathcal{Q} \hookrightarrow \mathbb{R}^m$. Not realised in the network.

Alignment. $D_{\text{ambient}} = m$, under which faithfulness reduces to C^1-approximation of ϕ.

B Proofs of Classical Results

This appendix collects the proofs for all results in the main text that apply classical mathematics without new ideas. None of the proofs in this appendix constitute a mathematical contribution of this paper.

B.1 Proof of Theorem 4.6 (Finite-Sample Traversal)

Classical source: Whyburn (1942); Rourke–Sanderson (1972), Ch. 1.

Proof. Because the points are pairwise distinct, there exists a vector $v \in \mathbb{R}^m$ such that the projections $\lambda_i = \langle v, x_i \rangle$ are pairwise distinct. The set of bad directions — those v for which some pair of projections coincide — is a finite union of proper hyperplanes $\{v : \langle v, x_i - x_j \rangle = 0\}$, which has Lebesgue measure zero; a generic v avoids all of them.

Relabel so that $\lambda_1 < \lambda_2 < \cdots < \lambda_n$. Define γ piecewise linearly on $[(i-1)/(n-1),\ i/(n-1)]$ by

$$\gamma\left(\frac{i-1+s}{n-1}\right) = (1-s) \cdot x_i + s \cdot x_{i+1}, \quad s \in [0,1].$$

This map is continuous by construction and passes through each x_i. For injectivity, consider $\phi(t) = \langle v, \gamma(t) \rangle$. On the i-th segment, $\phi((i-1+s)/(n-1)) = (1-s)\lambda_i + s\lambda_{i+1}$, which is strictly increasing in s because $\lambda_i < \lambda_{i+1}$. Hence ϕ is strictly increasing on $[0,1]$, so $t \neq t'$ implies $\gamma(t) \neq \gamma(t')$. □

B.2 Proof of Theorem 3.2 (Topological Embedding Floor)

Proof. By Definition 3.1, $\widetilde{E}$ topology-faithful means $\widetilde{E}$ is a homeomorphism onto $\widetilde{E}(\mathcal{Q}) \subset \mathbb{R}^{D_{\text{ambient}}}$, i.e., a topological embedding of $\mathcal{Q}$ into $\mathbb{R}^{D_{\text{ambient}}}$. By definition of $\mathrm{emb}_{\mathrm{top}}(\mathcal{Q})$ as the smallest d for which such an embedding exists, $D_{\text{ambient}} \geq \mathrm{emb}_{\mathrm{top}}(\mathcal{Q})$. □

B.3 Proof of Theorem 3.3 (Covering-Dimension Floor)

Classical sources: Brouwer (1911); Hurewicz–Wallman (1941); Engelking (1978), Theorem 3.1.2.

Proof. Suppose $\mathcal{Q}$ embeds topologically into $\mathbb{R}^d$ via some $\iota : \mathcal{Q} \hookrightarrow \mathbb{R}^d$. Covering dimension is a topological invariant (Engelking 1978, Theorem 1.7.7): homeomorphic spaces have equal covering dimension. It is also monotone for subspaces of metrisable spaces (Engelking 1978, Theorem 3.1.2): any subspace of $\mathbb{R}^d$ has covering dimension at most d. Therefore

$$\dim_{\mathrm{cov}}(\mathcal{Q}) = \dim_{\mathrm{cov}}(\iota(\mathcal{Q})) \leq \dim_{\mathrm{cov}}(\mathbb{R}^d) = d.$$

Taking the infimum over d admitting an embedding gives $\dim_{\mathrm{cov}}(\mathcal{Q}) \leq \mathrm{emb}_{\mathrm{top}}(\mathcal{Q})$, i.e., $\mathrm{emb}_{\mathrm{top}}(\mathcal{Q}) \geq \dim_{\mathrm{cov}}(\mathcal{Q}) = k$. □

Remark B.1 ($D_{\text{ambient}} = 1$, scalar bottleneck)*.* When $D_{\text{ambient}} = 1$, Theorem 3.2 combined with Theorem 3.3 gives $k \leq 1$ for any faithfully encoded compact connected quotient $\mathcal{Q}$. Combined with Munkres (2000), Theorems 24.1 and 27.3 — which characterise compact connected subsets of $\mathbb{R}$ as closed intervals — this gives $\mathcal{Q} \cong [a,b]$. Any quotient with nontrivial loop structure (S^1, T^k, S^k for $k \geq 2$) cannot be faithfully encoded by a scalar.

B.4 Proof of Lemma 3.4 (Quotient Embedding Requirement)

Proof. If $\widetilde{E}$ is topology-faithful, then by Definition 3.1 it is a homeomorphism from $\mathcal{Q}$ onto $\widetilde{E}(\mathcal{Q}) \subset \mathbb{R}^d$. A homeomorphism onto a subspace is by definition a topological embedding. The contrapositive gives: if $\mathcal{Q}$ is not embeddable in $\mathbb{R}^d$, no topology-faithful map $\mathcal{Q} \to \mathbb{R}^d$ exists. □

B.5 Proof of Corollary 3.6 (Below-Floor Collapse)

Proof. Immediate contrapositive of Theorem 3.2: if a topology-faithful map existed, we would have $D_{\text{ambient}} \geq \text{emb}_{\text{top}}(\mathcal{Q})$, contradicting $D_{\text{ambient}} < \text{emb}_{\text{top}}(\mathcal{Q})$. For compact Hausdorff $\mathcal{Q}$ and Euclidean target, a continuous injection is automatically a topological embedding (Munkres 2000, Theorem 26.6, applied to compact-domain continuous bijections onto Hausdorff codomains), so the failure modes "identifies distinct points" and "fails to preserve neighbourhood topology" collapse: non-embedding implies non-injectivity. □

B.6 Proof of Theorem 4.4 (Task-Map Collapse)

Classical sources for (i)–(ii): Lehmann–Scheffé (1950); Shamir, Sabato, and Tishby (2010).

Proof. **(i) Sufficiency.** By definition $E_{\text{label}}(x) = f(x)$, so the identity map recovers the target: $f = \text{id} \circ E_{\text{label}}$.

(ii) Minimality. Let $E : \mathcal{X} \to Z$ be any sufficient representation. Then there exists $g : Z \to Y$ with $f = g \circ E$. Since $E_{\text{label}}(x) = f(x)$ for all x, substituting gives $E_{\text{label}} = g \circ E$, which is exactly $E_{\text{label}} \preceq_{\text{comp}} E$ by definition with witness $h = g$. Since E was arbitrary, E_{label} is a lower bound of all sufficient representations. Because it is itself sufficient, it is the minimum element.

(iii) Failure to embed. Because $f = t \circ q$, the label representation factors as $E_{\text{label}} = t \circ q$, so the induced map on $\mathcal{Q}$ is $\widetilde{E_{\text{label}}} = t$. If t is not injective, there exist distinct $q_1, q_2 \in \mathcal{Q}$ with $t(q_1) = t(q_2)$, giving $\widetilde{E_{\text{label}}}(q_1) = \widetilde{E_{\text{label}}}(q_2)$. A topological embedding must be injective, so $\widetilde{E_{\text{label}}}$ is not a topological embedding of $\mathcal{Q}$. □

B.7 Proof of Proposition 7.1 (Topology–Geometry Gap)

Classical source: Burago, Burago, and Ivanov (2001), Chapter 1.

Proof. **Topology-faithfulness.** Each $\widetilde{E}_\alpha$ is strictly increasing and continuous with continuous inverse $x \mapsto x^{1/\alpha}$, so it is a homeomorphism of $[0, 1]$ onto itself.

Unbounded distortion. Take $x_\alpha = 0$, $y_\alpha = 1/\alpha$. The source distance is $d_{\mathcal{Q}}(x_\alpha, y_\alpha) = 1/\alpha$. The image distance is

$$|\widetilde{E}_\alpha(x_\alpha) - \widetilde{E}_\alpha(y_\alpha)| = (1/\alpha)^\alpha = \alpha^{-\alpha}.$$

The compression ratio is $\alpha^{-\alpha}/(1/\alpha) = \alpha^{1-\alpha} \to 0$ as $\alpha \to \infty$. □

C Novelty and Contribution Attribution

This appendix provides a unified account of (i) the novelty status of each formal result, and (ii) the intellectual contribution of each party to the paper. The two assessments are presented together because they address the same underlying question: what is genuinely new here, and who produced it?

The attribution table below uses the weight scheme from Appendix D: **0** (not applicable), + (substantive contribution), ++ (primary responsibility; this party could have produced this result independently). The **AI** ++ weight is reserved for results volunteered without being asked; executing a given direction is +, not ++.

Attribution table

Result / Contribution	Character	Lit.	Author	AI
Master frame				
Criterion silence as the master claim	novel framing	0	++	0
Two-regime split (Locatello vs. this paper)	novel framing	+	++	0
Constructive framework				
Embedding-space program (§1.3)	novel framework	0	++	0
Chain of objects with corrected $E \to E(\mathcal{X})$ mapping	novel formulation	0	++	0
Three architectural-adequacy conditions	novel formulation	0	++	0
Compression preorder (Definition 4.2)	novel formalisation	0	++	0
Quotient-respect failure (Definition 5.1)	novel formalisation	0	++	0
Two-subcase decomposition (refinement / incomparable)	novel formalisation	0	++	0
Lawful Abstraction Decomposition (Proposition 6.1)	novel framing	0	++	0
Cylinder / phase / sine running example	novel illustration	0	++	0
Theorems applied as tools from classical mathematics				
Theorem 4.6 (Finite-Sample Traversal)	classical folklore	++	+	0
Theorem 3.2 (Topological Embedding Floor)	definitional	++	0	0
Theorem 3.3 (Covering-Dimension Floor)	classical dim. theory	++	+	0
Lemma 3.4 (Quotient Embedding Req.)	tautological reframe	++	0	0
Corollary 3.6 (Below-Floor Collapse)	immediate consequence	++	0	0
Theorem 4.4 (i)–(ii) Minimal Sufficiency	classical statistics	++	+	0
Lemma 4.13 (Tietze extension)	classical analysis	++	+	0
Proposition 7.1 (Topology–Geometry Gap)	classical metric geom.	++	+	0
Novel theorems and connections				
Theorem 4.4(iii) Task-Map Collapse	genuinely new	+	++	0
Theorem 4.7 (Graded Simplicial Spans)	genuinely new	+	++	0
Corollary 4.9 (lower bound $N_k(n)$)	novel observation	0	++	0
Asymmetry of quotient-respect (§5.1)	novel observation	0	++	0
Incomparable subcase identification	novel observation	0	++	0
Locatello as parallel-manifestation (§6.3)	novel framing	+	++	0
Kvalheim–Sontag subsumption acknowledgement (Remark 3.5)	novel framing	+	++	0
Scientific direction and editorial decisions				
Decision to replace AI proofs with classical citations	scientific direction	0	++	0
Novelty classification of all theorems	scientific direction	0	++	0
Reference selection with interpretive framing	scientific direction	0	++	0
Restructure to embedding-space program rewrite	scientific direction	0	++	0
AI-assisted execution (all under author direction)				

C.1 Primary contribution

The paper's primary contribution is the master frame — predictive loss does not certify quotient faithfulness, regime-independently — together with the embedding-space program that gives the structural account of what faithfulness requires, and three logically independent failure modes that manifest criterion silence in the regime where the admissibility relation has been declared. The two formalisations introduced — the compression preorder and the quotient-respect failure asymmetry — give precise language for the gap. The individual mathematical theorems include classical results applied in a new context (Theorems 4.6, 3.3, Lemma 3.4, parts (i)–(ii) of Theorem 4.4, Lemma 4.13, Proposition 7.1) and genuinely new results (Theorem 4.4(iii), Theorem 4.7, Definition 5.1 with its two-subcase analysis). The genuinely novel intellectual content is the framework itself, the manifestation reading of the failure modes, and the connections drawn between classical topology, statistical sufficiency, and representation learning evaluation.

C.2 Critical prior work

The three most important prior works are: Locatello et al. [2019], whose unidentifiability result for disentanglement is reframed in this paper as the parallel manifestation of criterion silence in the $\sim$-undeclared regime; Tishby et al. [1999], Tishby and Zaslavsky [2015], whose information bottleneck is the specific compression objective shown to actively destroy quotient structure when the task map is non-injective; and Kvalheim and Sontag [2024], whose topological results on autoencoders are strictly stronger than Lemma 3.4 and must be engaged prominently.

C.3 AI tools

Tool	Version	Phase	Role
ChatGPT	GPT-4 era	Ideation	Exploratory conversation partner
GPT-5.4	OpenAI, Mar. 2026	Proof drafts	Source proof drafts (superseded by author)
Claude Sonnet 4.6	Anthropic, Apr. 2026	Writing	Prose drafting; LaTeX typesetting; all under author direction
Claude Opus 4.7	Anthropic, May 2026	Restructure pass	Embedding-space program rewrite, framing patches, integration; all under author direction

Generative AI outputs are non-deterministic and cannot be exactly reproduced. The paper should be evaluated on its intellectual and mathematical content alone. The intellectual genesis of all novel contributions belongs to the first author; the AI's role was expressive, not generative. The first author takes full scientific and intellectual responsibility for all claims.

D The Attribution Prompt and Worked Example

D.1 The prompt

The following prompt can be used to generate a contribution attribution table for any AI-assisted academic paper. It is designed to be honest and reproducible.

> Read the full paper and the full conversation history before making any attributions. The paper alone cannot establish who originated an idea or whether the AI acted on a direction or volunteered something unprompted. The conversation history is the primary evidence for Author vs. AI weight assignments. Attributions made without it are unreliable.
>
> Identify every distinct intellectual unit: named theorems, definitions, conjectures, remarks, open problems, syntheses of existing results, reframings of existing phenomena, and novel connections. Each is a separate row. Do not include purely pedagogical illustrations.
>
> For each row, write a short descriptive label in the Character column — one to four words, honest and descriptive. A row that is genuinely two things can say so.
>
> Assign weights across three columns — **Lit.** (prior literature), **Author**, **AI** — using:
>
> - **0** not applicable
> - + substantive contribution
> - ++ primary responsibility; this party could have produced this result independently
>
> *Calibration:* Author ++ when the idea originated with the human author and would not have appeared without their specific input. AI ++ reserved exclusively for results volunteered without being asked — not for executing a given direction, not for writing prose, not for identifying a theorem when prompted. Supplying a theorem when asked is +. Proposing its relevance unprompted is ++.
>
> Produce the table, then write: a **Primary Contribution** paragraph stating the paper's main contribution plainly; a **Critical Prior Work** subsection naming the two or three papers the work builds on most directly; a check for any conflation between existence results and constructive or learnability results; and an **AI Tools** subsection listing each tool, its version, phase, and role, closing with a statement that the human author takes full responsibility for all claims.

D.2 Worked example: this paper

The table in Appendix C was produced by applying the prompt above to Paper I after reading the full drafting conversation. The AI ++ column is empty throughout: no row in this paper meets the criterion of an unsolicited contribution that the AI volunteered without being asked. Every AI contribution was in response to a specific direction from the author. This is the correct result for a paper in which the author provided detailed mathematical outlines, directed the proof strategy, selected all references, and made the decision to replace AI-generated proofs with classical citations. The closest candidates —

the prose formulation of the restructure to the embedding-space program, the corrected $E \to E(\mathcal{X})$ mapping, and the asymmetry of the quotient-respect condition — were all responses to explicit author prompts and corrections, and so receive + rather than ++. Any attribution of ++ to the AI in this paper would misrepresent the collaboration.